

MCQ & SHORT QUESTION ANSWERS – NOTES**BY : DR. SNEHAL K JOSHI**

1. The sutra “Nikhilam Navatashcaramam Dashatah” is mainly used for:

- A) Squaring numbers ending in 5
- B) Multiplying numbers close to 10, 100, 1000
- C) Finding square roots
- D) Solving linear equations

 **Correct Answer: B**

2. Using Nikhilam Sutra, what is 98×97 ?

- A) 9506
- B) 9606
- C) 9406
- D) 9706

 **Correct Answer: B**

($98 = 100 - 2$, $97 = 100 - 3 \rightarrow 100 - 5 = 95 \mid 2 \times 3 = 06 \rightarrow 9506 \rightarrow$ correction:
Actually $98 \times 97 = 9506$, so correct option is A)

✓ **Correct Answer: A**

3. The sutra Ekadhikena Purvena is best suited for:

- A) Division by 9
- B) Squaring numbers
- C) Multiplication of numbers ending in 5
- D) Cube root extraction

 **Correct Answer: C**

4. Using Ekadhikena Purvena, what is 35^2 ?

- A) 1225
- B) 1125
- C) 1325
- D) 1025

 **Correct Answer: A**

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5. The sutra Udharan is used for:

- A) Multiplication
- B) Subtraction
- C) Extraction of roots
- D) Addition

 **Correct Answer: C**

6. The sutra Paraavartya is mainly applied for:

- A) Fast multiplication
- B) Fast division
- C) Squaring
- D) Factorization

 **Correct Answer: B**

7. The sutra Shunyam Saamyasamuccaye is used when:

- A) Products are equal
- B) Differences are equal
- C) Sums on both sides are equal
- D) Quotients are equal

 **Correct Answer: C**

8. If $2x + 7 = x + 12$, using Shunyam Saamyasamuccaye, the value of x is:

- A) 3
- B) 4
- C) 5
- D) 6

 **Correct Answer: C**

9. Using Nikhilam Sutra, what is 104×98 ?

- A) 10192
- B) 10292
- C) 10092
- D) 10392

✓ **Correct Answer: B**

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10. The literal meaning of Paraavartya Yojayet is:

- A) By observation
- B) Transpose and apply
- C) By addition and subtraction
- D) By splitting

✓ **Correct Answer: B**

11. The sutra Anurupyena is mainly used for:

- A) Fast subtraction
- B) Proportional calculations
- C) Squaring numbers
- D) Root extraction

✓ **Correct Answer: B**

12. The meaning of Anurupyena is:

- A) By completion
- B) By parts and whole
- C) Proportionately
- D) By motion

✓ **Correct Answer: C**

13. The sutra Sankalana–Vyavakalanabhyam is used for:

- A) Only multiplication
- B) Only division

- C) By addition and subtraction
- D) By squaring

✓ **Correct Answer: C**

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14. The literal meaning of Sankalana–Vyavakalanabhyam is:

- A) By observation
- B) By proportion
- C) By completion
- D) By addition and by subtraction

✓ **Correct Answer: D**

15. The sutra Puranapuranabhyam is mainly applied for:

- A) Completing or non-completing values
- B) Finding square roots
- C) Fast division
- D) Subtraction only

✓ **Correct Answer: A**

16. The sutra Chalana–Kalana is used for:

- A) Static calculations
- B) Motion-based shifting method
- C) Squaring numbers
- D) Root extraction

✓ **Correct Answer: B**

17. The meaning of Chalana–Kalana is:

- A) By parts
- B) By proportion
- C) By motion or shift
- D) By completion

✓ **Correct Answer: C**

18. The sutra Yavadunam refers to:

- A) Whatever is the deficiency
- B) By addition
- C) The product of sum
- D) The remainder

 **Correct Answer: A**

MCQ & SHORT QUESTION ANSWERS – NOTES

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19. The sutra Vyastisamanstih represents:

- A) Product of sum
- B) Remainder
- C) Proportion
- D) Parts and the whole

 **Correct Answer: D**

20. The meaning of Puranapurabhyam is:

- A) By motion
- B) By transposition
- C) By completion or non-completion
- D) By addition

 **Correct Answer: C**

21. Which sutra is best suited for solving proportion-based problems quickly?

- A) Yavadunam
- B) Anurupyena
- C) Chalana–Kalana
- D) Vyastisamanstih

 **Correct Answer: B**

22. The sutra Sankalana–Vyavakalanabhyam is useful when calculations involve:

- A) Only division
- B) Only multiplication
- C) Addition and subtraction together
- D) Square roots

✓ **Correct Answer: C**

MCQ & SHORT QUESTION ANSWERS – NOTES

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23. Which sutra is applied when a value is less than a known base?

- A) Anurupyena
- B) Vyastisamanstih
- C) Chalana–Kalana
- D) Yavadunam

✓ **Correct Answer: D**

24. The meaning of Vyastisamanstih is:

- A) The remainder
- B) Product of sum
- C) Parts and the whole
- D) By proportion

✓ **Correct Answer: C**

25. Which sutra uses the concept of shifting digits during calculation?

- A) Chalana–Kalana
- B) Puranapurabhyam
- C) Anurupyena
- D) Yavadunam

✓ **Correct Answer: A**

26. The sutra Puranapurabhyam focuses on:

- A) Transposition
- B) Completion and non-completion

- C) Addition
- D) Deficiency

✓ **Correct Answer: B**

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27. Which sutra literally means “By motion or applying a shift”?

- A) Anurupyena
- B) Vyastisamanstih
- C) Chalana–Kalana
- D) Sankalana–Vyavakalanabhyam

✓ **Correct Answer: C**

28. When numbers are in a definite ratio, which sutra is applied?

- A) Yavadunam
- B) Puranapurabhyam
- C) Sankalana–Vyavakalanabhyam
- D) Anurupyena

✓ **Correct Answer: D**

29. The sutra Yavadunam is mainly used for:

- A) Proportional increase
- B) Completing values
- C) Deficiency from a base
- D) Finding remainder

✓ **Correct Answer: C**

30. Which sutra explains the relationship between individual components and the complete value?

- A) Vyastisamanstih
- B) Chalana–Kalana
- C) Anurupyena
- D) Sankalana–Vyavakalanabhyam

✓ **Correct Answer: A**

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31. The sutra Sesanyan is used to find:

- A) The product
- B) The quotient
- C) The remainder
- D) The square root

✓ **Correct Answer: C**

32. The meaning of Gunitasamuchyah is:

- A) The sum of the product
- B) The product of the sum
- C) The remainder
- D) The expanded form

✓ **Correct Answer: B**

33. Which sutra is applied when an algebraic expression needs to be expanded?

- A) Rupan
- B) Sesanyan
- C) Vistaran
- D) Chidana

✓ **Correct Answer: C**

34. The sutra Rupan mainly deals with:

- A) Splitting numbers
- B) Form and structure of expressions
- C) Remainders
- D) Expansion

✓ **Correct Answer: B**

35. The sutra Chidana means:

- A) Expansion
- B) Completion
- C) Splitting
- D) Remainder

✓ **Correct Answer: C**

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36. Which sutra helps in checking divisibility using remainders?

- A) Gunitasamuchyah
- B) Sesanyan
- C) Vistaran
- D) Rupan

✓ **Correct Answer: B**

37. Which sutra is useful for factorizing expressions?

- A) Rupan
- B) Sesanyan
- C) Gunitasamuchyah
- D) Vistaran

✓ **Correct Answer: A**

38. The sutra Vistaran is helpful when:

- A) Calculating remainder
- B) Writing in standard form
- C) Expanding algebraic expressions
- D) Splitting numbers

✓ **Correct Answer: C**

39. Which sutra can be applied to break a complex number into simpler parts?

- A) Sesanyan
- B) Rupan
- C) Gunitasamuchyah
- D) Chidana

✓ **Correct Answer: D**

MCQ & SHORT QUESTION ANSWERS – NOTES

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40. The sutra Gunitasamuchyah is useful for:

- A) Finding remainder
- B) Checking multiplication
- C) Expanding expressions
- D) Forming equations

✓ **Correct Answer: B**

41. Which sutra is useful to determine the remainder of a number without full division?

- A) Vistaran
- B) Chidana
- C) Sesanyan
- D) Rupan

✓ **Correct Answer: C**

42. The sutra Gunitasamuchyah helps in:

- A) Finding the remainder
- B) Verifying multiplication
- C) Expanding expressions
- D) Splitting numbers

✓ **Correct Answer: B**

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43. Which sutra is applied when an expression is written in expanded polynomial form?

- A) Sesanyan
- B) Vistaran
- C) Rupan
- D) Chidana

 **Correct Answer: B**

44. The word Rupan in Vedic Mathematics means:

- A) Remainder
- B) Completion
- C) Form
- D) Product

 **Correct Answer: C**

45. Which sutra allows breaking down a difficult problem into simpler steps?

- A) Vistaran
- B) Gunitasamuchyah
- C) Sesanyan
- D) Chidana

 **Correct Answer: D**

46. Which sutra can be used to check if a number is divisible by another?

- A) Sesanyan
- B) Rupan
- C) Vistaran
- D) Chidana

 **Correct Answer: A**

47. Which sutra focuses on the structure of mathematical expressions?

- A) Chidana
- B) Sesanyan
- C) Gunitasamuchyah
- D) Rupan

✓ **Correct Answer: D**

48. The sutra Vistaran is used mainly for:

- A) Factoring
- B) Expansion
- C) Remainders
- D) Proportions

✓ **Correct Answer: B**

49. The sutra Chidana helps in:

- A) Combining terms
- B) Splitting complex values
- C) Finding products
- D) Calculating remainder

✓ **Correct Answer: B**

50. Which sutra verifies multiplication results?

- A) Sesanyan
- B) Vistaran
- C) Rupan
- D) Gunitasamuchyah

✓ **Correct Answer: D**

SHORT QUESTION ANSWERS – NOTES

BY : DR. SNEHAL K JOSHI

1. Explain the meaning and application of the sutra Nikhilam Navatashcaramam Dashatah.

Answer:

This sutra means “All from 9 and the last from 10.” It is mainly used for fast multiplication of numbers close to base 10, 100, 1000, etc. The method reduces calculation steps by using complements. It makes mental multiplication quick and easy.

2. What is the principle behind Ekadhikena Purvena? Where is it commonly used?

Answer:

Ekadhikena Purvena means “By one more than the previous one.” It is commonly used for squaring numbers ending in 5 and for decimal multiplication. The method simplifies calculations by increasing the previous digit by one and multiplying.

3. Describe the sutra Udharan and its use in extraction methods.

Answer:

Udharan means “The extraction.” It is used in extracting roots such as square roots and cube roots. The sutra provides a systematic and quick method to find roots mentally. It reduces long traditional steps.

4. What is meant by Paraavartya? How does it help in simplification?

Answer:

Paraavartya means “Transposition and cancellation.” It is mainly used in division of algebraic expressions and equations. The method simplifies problems by rearranging terms and cancelling common factors quickly.

5. Explain the concept of Shunyam Saamyasamuccaye.

Answer:

This sutra means “When the sum is the same, that sum is zero.” It is applied in solving algebraic equations where equal sums on both sides cancel each other. It helps in reducing lengthy equations into simpler forms.

6. What is the meaning of Anurupyena and how is it applied?**Answer:**

Anurupyena means “Proportionately.” It is used when numbers are in proportion to a base number. The sutra helps in fast multiplication and division by adjusting values proportionally to a known base.

7. Describe the working principle of Sankalana–Vyavakalanabhyam.**Answer:**

This sutra means “By addition and by subtraction.” It is used in solving equations and simplifying algebraic expressions. The method combines addition and subtraction techniques to reach the solution faster.

8. Explain the idea behind Puranapuranabhyam.**Answer:**

Puranapuranabhyam means “By the completion or non-completion.” It is useful in algebraic factorization and solving quadratic equations. The sutra works by completing the required values to simplify calculations.

9. What is the role of Chalana–Kalana in fast calculations?**Answer:**

Chalana–Kalana means “By motion or applying a shift.” It is used in multiplication and division involving place value shifting. This sutra makes calculations faster by moving digits instead of rewriting numbers.

10. Explain the sutra Yavadunam and its practical importance.**Answer:**

Yavadunam means “Whatever is the deficiency.” It is used for fast multiplication of numbers slightly less than a base. The deficiency from the base is used to compute the result quickly and accurately.

11. Explain the importance of the sutra Vyastisamanstih in understanding algebraic expressions.**Answer:**

Vyastisamanstih means “The parts and the whole.” It explains how individual terms combine to form a complete expression. This sutra helps students understand polynomial identities and factorization. It builds clarity in algebraic reasoning.

12. How does the sutra Sesanyan help in divisibility and verification?**Answer:**

Sesanyan means “The remainder.” It helps in finding remainders quickly without performing full division. This sutra is useful in checking divisibility and verifying the correctness of calculations. It saves time and effort.

13. Explain the checking principle of the sutra Gunitasamuchyah.**Answer:**

Gunitasamuchyah means “The product of the sum.” It is used as a verification method for multiplication. The product should match the sum of the product of digits. It helps in detecting calculation errors.

14. Describe the role of Vistaran in algebraic operations.**Answer:**

Vistaran means “Expansion.” It is used to expand algebraic expressions like $(a+b)^2$ or $(x+y)(x-y)$. The sutra provides a systematic method to expand terms correctly and quickly.

15. What is the purpose of the sutra Rupan in mathematical simplification?**Answer:**

Rupan means “Form.” It helps in converting expressions into suitable forms for simplification. This sutra is useful in factorization and equation solving. It improves clarity and structure in problem-solving.

16. Explain how Chidana helps in mental calculations.**Answer:**

Chidana means “By splitting.” It simplifies complex numbers by breaking them

into smaller parts. This makes multiplication and division easier to perform mentally. It improves speed and accuracy.

17. How does Vyastisamanstih support factorization techniques?

Answer:

This sutra explains the relationship between individual terms and the whole expression. It helps in identifying common factors and applying identities. It is useful in simplifying polynomial expressions.

18. Why is Sesanyan useful in competitive examinations?

Answer:

Sesanyan allows quick calculation of remainders. Many aptitude and competitive exam problems are based on divisibility and remainders. This sutra saves time and improves calculation speed.

19. How is Vistaran helpful in solving quadratic expressions?

Answer:

Vistaran expands expressions like $(x+3)(x+5)$ into quadratic form. It helps in forming standard equations. This makes solving quadratic problems systematic and easier.

20. Explain why Chidana is considered a powerful Vedic Mathematics technique.

Answer:

Chidana simplifies problems by splitting numbers into manageable parts. It reduces long calculation steps. This makes it ideal for fast mental math and competitive problem-solving.

21. Explain the practical importance of the sutra Anurupyena (अनुप्रयेन) in daily calculations.

Answer:

Anurupyena (अनुप्रयेन) means “Proportionately.” It is used when numbers are related proportionally to a known base. This sutra helps in quick multiplication and division by adjusting values in proportion. It simplifies large calculations.

22. How does Sankalana–Vyavakalanabhyam (संकलन-व्यवकलनाभ्याम्) simplify algebraic equations?

Answer:

Sankalana–Vyavakalanabhyam (संकलन-व्यवकलनाभ्याम्) means “By addition and by subtraction.” It is used to simplify equations by systematically adding or subtracting terms. It reduces complex expressions into simpler forms and speeds up problem solving.

23. Explain the application of Puranapurabhyam (पूर्णापूर्णाभ्याम्) in quadratic equations.

Answer:

Puranapurabhyam (पूर्णापूर्णाभ्याम्) means “By the completion or non-completion.” It is useful in completing the square while solving quadratic equations. This method helps in forming perfect square expressions for easy solutions.

24. Describe how Chalana–Kalana (चलन-कलन) helps in place-value based calculations.

Answer:

Chalana–Kalana (चलन-कलन) means “By motion or by applying a shift.” It is used in multiplication and division by shifting digits according to place value. This makes calculations faster and avoids lengthy written work.

25. How does Yavadunam (यावदूनम्) improve speed in multiplication near a base?

Answer:

Yavadunam (यावदूनम्) means “Whatever is the deficiency.” It is used when numbers are slightly less than a base like 10, 100, or 1000. The deficiency is used to calculate the result quickly. It saves time and effort.

26. Explain the use of Nikhilam Navatashcaramam Dashatah (निखिलं नवतश्चरमम् दशतः) in mental mathematics.

Answer:

Nikhilam Navatashcaramam Dashatah (निखिलं नवतश्चरमम् दशतः) means “All from 9 and the last from 10.” It is used for fast multiplication of numbers close to a base. The method uses complements, making mental calculation very easy and fast.

27. How does Ekadhikena Purvena (एकाधिकेन पूर्वेण) help in squaring numbers ending in 5?

Answer:

Ekadhikena Purvena (एकाधिकेन पूर्वेण) means “By one more than the previous one.” It is used to quickly square numbers ending in 5. The previous digit is multiplied by one more than itself, followed by 25.

28. What role does Udharan (उद्धारण) play in root extraction?

Answer:

Udharan (उद्धारण) means “The extraction.” It is used to extract square roots and cube roots. The sutra provides a systematic and faster method than traditional approaches. It improves mental calculation skills.

29. How does Paraavartya (परावर्त्य) help in algebraic division?

Answer:

Paraavartya (परावर्त्य) means “Transposition and cancellation.” It is used to divide algebraic expressions easily. The method simplifies division by rearranging and cancelling terms.

30. Explain the importance of Shunyam Saamyasamuccaye (शून्यं साम्यसमुच्चयै) in equation solving.

Answer:

Shunyam Saamyasamuccaye (शून्यं साम्यसमुच्चयै) means “When the sum is the same, that sum is zero.” It is used when identical expressions appear on both sides of an equation. It simplifies the equation by cancelling equal terms.

31. Explain the meaning and application of Vyastisamanstih (व्यस्तिसमंस्थि:).**Answer:**

Vyastisamanstih (व्यस्तिसमंस्थि:) means “The parts and the whole.” It helps in understanding how individual terms relate to the total expression. The sutra is used in algebra to simplify polynomials and in verifying multiplication results. It provides clarity in combining and splitting terms.

32. How does Sesanyan (शेषान्) help in checking calculations?**Answer:**

Sesanyan (शेषान्) means “The remainder.” It is used to find remainders quickly without performing full division. This helps verify multiplication, division, and algebraic operations efficiently. It is useful in both mental math and competitive exams.

33. Describe the practical use of Gunitasamuchyah (गुणितसमुच्चय:).**Answer:**

Gunitasamuchyah (गुणितसमुच्चय:) means “The product of the sum.” It is used to check multiplication results by comparing the sum of digits with the product of sums. This sutra is a quick method to detect errors in calculations.

34. How is Vistaran (विस्तारण) applied in algebraic expansion?**Answer:**

Vistaran (विस्तारण) means “Expansion.” It is used to expand expressions like $(a+b)^2$ or $(x+y)(x-y)$ systematically. The sutra ensures accurate multiplication of terms and helps in solving algebraic problems efficiently.

35. Explain the significance of Rupan (रूपन्) in simplification of expressions.**Answer:**

Rupan (रूपन्) means “Form.” It helps to convert numbers or expressions into a suitable form for easier calculation. This sutra is applied in factorization, equation solving, and simplifying complex algebraic expressions.

36. What is the purpose of Chidana (छिदन) in mental mathematics?**Answer:**

Chidana (छिदन) means “By splitting.” It simplifies complex calculations by breaking numbers into smaller, manageable parts. This technique is particularly useful in multiplication and division, making mental math faster and more accurate.

37. How does Anurupyena (अनुप्रयेन) help in proportional calculations?**Answer:**

Anurupyena (अनुप्रयेन) means “Proportionately.” It is used when numbers are in proportion to a base or reference value. This sutra simplifies multiplication and division by adjusting numbers proportionally, reducing calculation time.

38. Explain the role of Sankalana–Vyavakalanabhyam (संकलन-व्यवकलनाभ्याम्) in solving equations.**Answer:**

Sankalana–Vyavakalanabhyam (संकलन-व्यवकलनाभ्याम्) means “By addition and by subtraction.” It helps in simplifying equations by combining terms through addition or subtraction. The sutra reduces the complexity of algebraic operations.

39. Describe how Puranapurabhyam (पूर्णापूर्णाभ्याम्) aids in factorization.**Answer:**

Puranapurabhyam (पूर्णापूर्णाभ्याम्) means “By completion or non-completion.” It is used to factorize numbers and algebraic expressions

efficiently. By completing or adjusting numbers, the sutra simplifies calculations and reduces errors.

40. How is Chalana–Kalana (चलन-कलन) used in digit-shifting methods?

Answer:

Chalana–Kalana (चलन-कलन) means “By motion or by applying a shift.” It is applied in multiplication and division by shifting digits according to place value. This method speeds up calculations and avoids lengthy written procedures.

41. Explain the application of Yavadunam (यावदूनम्) in multiplication near base numbers.

Answer:

Yavadunam (यावदूनम्) means “Whatever is the deficiency.” It is used when numbers are slightly less than a base like 10, 100, or 1000. By using the deficiency, multiplication can be done quickly. This method simplifies calculations and improves speed.

42. How does Nikhilam Navatashcaramam Dashatah (निखिलं नवतश्चरमम् दशतः) aid in fast multiplication?

Answer:

Nikhilam Navatashcaramam Dashatah (निखिलं नवतश्चरमम् दशतः) means “All from 9 and the last from 10.” It is used to multiply numbers close to a power of 10. The sutra uses complements to simplify calculations, enabling quick mental computation.

43. Describe how Ekadhikena Purvena (एकाधिकेन पूर्वेण) is used to square numbers ending in 5.

Answer:

Ekadhikena Purvena (एकाधिकेन पूर्वेण) means “By one more than the previous one.” For numbers ending in 5, the previous digit is multiplied by one more than itself, and 25 is appended. This method provides a fast and accurate way to square such numbers.

44. Explain the role of Udharan (उद्धारण) in root extraction.**Answer:**

Udharan (उद्धारण) means “The extraction.” It helps in finding square roots and cube roots efficiently. The sutra provides a systematic method, making calculations faster and reducing the chance of errors.

45. How does Paraavartya (परावर्त्य) simplify division problems?**Answer:**

Paraavartya (परावर्त्य) means “Transposition and cancellation.” It helps in dividing numbers or expressions by rearranging terms and cancelling common factors. This sutra simplifies the process and makes division faster.

46. Explain the significance of Shunyam Saamyasamuccaye (शून्यं साम्यसमुच्चयै) in solving equations.**Answer:**

Shunyam Saamyasamuccaye (शून्यं साम्यसमुच्चयै) means “When the sum is the same, that sum is zero.” It is applied when equal terms appear on both sides of an equation. By cancelling equal sums, solving the equation becomes easier and quicker.

47. How is Vistarana (विस्तारण) applied in polynomial expansion?**Answer:**

Vistarana (विस्तारण) means “Expansion.” It is used to multiply and expand algebraic expressions such as $(x+y)^2$ or $(a+b)(a-b)$. This sutra ensures all terms are included systematically and reduces errors in expansion.

48. What is the use of Rupan (रूपन्) in simplifying expressions?**Answer:**

Rupan (रूपन्) means “Form.” It helps convert numbers or algebraic expressions

into a convenient form for calculation. This sutra is particularly useful in factorization, equation solving, and simplification of complex expressions.

49. Describe the application of Chidana (छिदन) in mental calculations.

Answer:

Chidana (छिदन) means “By splitting.” It breaks numbers into smaller, manageable parts for multiplication or division. This technique improves speed and accuracy in mental calculations and reduces errors.

50. Explain how Anurupyena (अनुप्रयेन) is used for proportional multiplication.

Answer:

Anurupyena (अनुप्रयेन) means “Proportionately.” When numbers are in proportion to a base, this sutra allows quick multiplication or scaling. It simplifies calculations by maintaining proportional relationships, saving time and effort.

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